# C1.2 Cell respiration (HL)

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 40 |

**About 40 minutes,** or one section at a time. The number of marks shows how many separate points to make.

## Section A: Recall

10 MARKS

1 **DEFINE** oxidation in terms of hydrogen. [C1.2.7] [1]

2 **STATE** the net yield of ATP and of reduced NAD from glycolysis of one glucose. [C1.2.8] [2]

3 **STATE** the two products of anaerobic respiration in yeast. [C1.2.10] [2]

4 **NAME** the 4-carbon and 6-carbon compounds of the Krebs cycle. [C1.2.12] [2]

5 **STATE** where the electron transport chain is found. [C1.2.14] [1]

6 **NAME** the enzyme that makes ATP as protons flow back into the matrix. [C1.2.15] [1]

7 **STATE** the product formed when oxygen accepts electrons and protons. [C1.2.16] [1]

## Section B: Understanding

14 MARKS

8 **EXPLAIN** why muscle cells convert pyruvate to lactate when oxygen is short. [C1.2.9] [3]

9 **OUTLINE** the link reaction. [C1.2.11] [3]

10 **EXPLAIN** how reduced NAD leads to a proton gradient across the inner mitochondrial membrane. [3]
[C1.2.13, C1.2.14]

11 **EXPLAIN** why lipids release more energy per gram than carbohydrates. [C1.2.17] [2]

12 **STATE** why lipids cannot be used in anaerobic respiration. [C1.2.17] [1]

13 **EXPLAIN** why glycolysis begins by using ATP. [C1.2.8] [2]

## Section C: Working with evidence

10 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

**14** [C1.2.14, C1.2.15, C1.2.16]

Mitochondria were isolated from liver cells and given a respiratory substrate. Oxygen uptake and ATP production were measured under four conditions. DNP is a substance that makes the inner membrane leaky to protons.

| CONDITION | OXYGEN UPTAKE / NMOL MIN$^{-1}$ | ATP MADE / % OF CONDITION 2 |
|---|---|---|
| 1 SUBSTRATE ONLY, NO ADP | 8 | 0 |
| 2 SUBSTRATE AND ADP | 42 | 100 |
| 3 SUBSTRATE, ADP AND CYANIDE | 1 | 0 |
| 4 SUBSTRATE, ADP AND DNP | 60 | 2 |

**a** EXPLAIN why adding ADP increased oxygen uptake. [2]

**b** EXPLAIN the results with cyanide. [2]

**c** EXPLAIN why oxygen uptake was high but almost no ATP was made with DNP. [3]

**d** DNP was once sold as a slimming drug, but it causes dangerous overheating. SUGGEST why. [2]

**e** SUGGEST a substrate that could be given to isolated mitochondria. [1]

## Section D: Longer answer

6 MARKS

**15** EXPLAIN how energy from reduced NAD is used to make ATP in the mitochondrion. [C1.2.13–16] [6]